Extra Bite #14: Digital Logic

Gates, flip-flops, counters, multivibrators, truth tables, and positive logic conventions.

Group E7A · 11 pool questions · Standalone · 5-10 minute presentation · Practice exams at eham.net/exams

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Logic Gates: The Basic Verbs

Digital circuits are built from a handful of gate types. Each gate has 1 or 2 inputs and produces a 1 or 0 output based on a simple rule.

Gate	Rule	Q
OR	Output is **1 if any input is 1** (E7A08). Logical “or.”	E7A08
NAND	Output is **0 only if all inputs are 1** (E7A07). NOT-AND: the AND function inverted.	E7A07
XNOR	Two-input exclusive NOR: output **0 if one and only one input is 1** (E7A09). Stated positively: output is 1 when both inputs match.	E7A09

Truth table: a list of inputs and corresponding outputs for a digital device (E7A10). It is the contract that a gate or circuit advertises.

Positive Logic Convention

Positive logic: high voltage represents a 1, low voltage a 0 (E7A11). The opposite convention exists (“negative logic”) but the test wants positive.

Flip-Flops And Frequency Division

A **flip-flop** is the simplest **bistable** circuit (E7A01) — it stays in one of two stable states until it gets a clock pulse, then flips to the other. That switch-on-clock behavior makes it a halver: two input pulses produce one output cycle.

Question	Answer	Q
Circuit that is bistable	**A flip-flop** (E7A01).	E7A01
Divide a pulse train by 2	**A flip-flop** (E7A03). One toggle per input edge → output runs at half the input frequency.	E7A03
Flip-flops needed to divide by 16	**4 flip-flops** (E7A04). Each flip-flop divides by 2, and 2^4 = 16.	E7A04
Function of a decade counter	**Produces one output pulse for every 10 input pulses** (E7A02). A divide-by-10 counter.	E7A02

Multivibrators

Multivibrators are oscillator-like circuits built from two transistors (or gates) with cross-coupled feedback. They come in three flavors based on how many stable states they have.

Type	Behavior	Q
Astable	**Continuously alternates between two states without an external clock signal** (E7A05). A square-wave oscillator.	E7A05
Monostable	**Switches temporarily to an alternate state for a set time** (E7A06). A one-shot pulse generator.	E7A06
Bistable	Stays in either state indefinitely until clocked. Same family as a flip-flop.	—

Mental Cheat Sheet

- OR gate: 1 if any input is 1.
- NAND gate: 0 only if all inputs are 1.
- XNOR (2-input): 0 if exactly one input is 1; 1 when inputs match.
- Truth table = input → output for a digital device.
- Positive logic: high = 1, low = 0.
- Flip-flop = bistable.
- One flip-flop divides by 2.
- Divide by 16 → 4 flip-flops.
- Decade counter = divide by 10.
- Astable multivibrator: free-running square wave.
- Monostable: one-shot for a set time.

Common Traps

- NAND is 0 **only when all inputs are 1**, not “if some but not all inputs are 1.”
- XNOR outputs 0 when **exactly one** input is 1; the trap answers say “all 0” or “all 1.”
- Bistable = flip-flop, **not** AND/OR gates or bipolar amps.
- Decade counter divides by 10 (not by 100, not “decodes a decimal”).
- Astable runs **without** an external clock; monostable needs a trigger and times out.
- Positive logic is just the high-voltage convention, not a “logic state of truth.”

Takeaway: Gates do logic, flip-flops store one bit and halve frequency, decade counters divide by ten, multivibrators come in astable (free-running), monostable (one-shot), and bistable (latch) flavors. Truth tables describe everything; positive logic is the default.

Bite 14 Question Bank (E7A)

E7A01 Which circuit is bistable?

A. An AND gate.

E7A02. What is the function of a decade counter?
- A. It produces one output pulse for every 10 input pulses.
- B. It decodes a decimal number for display on a seven-segment LED display.
- C. It produces 10 output pulses for every input pulse.
- D. It decodes a binary number for display on a seven-segment LED display.

E7A03. Which of the following can divide the frequency of a pulse train by 2?
- A. An XOR gate.
- B. A flip-flop.
- C. An OR gate.
- D. A multiplexer.

E7A04. How many flip-flops are required to divide a signal frequency by 16?
- A. 4.
- B. 6.
- C. 8.
- D. 16.

E7A05. Which of the following circuits continuously alternates between two states without an external clock signal?
- A. Monostable multivibrator.
- B. J-K flip-flop.
- C. T flip-flop.
- D. Astable multivibrator.

E7A06. What is a characteristic of a monostable multivibrator?
- A. It switches temporarily to an alternate state for a set time.
- B. It produces a continuous square wave.
- C. It stores one bit of data.
- D. It maintains a constant output voltage, regardless of variations in the input voltage.

E7A07. What logical operation does a NAND gate perform?
- A. It produces a 0 at its output only if all inputs are 0.
- B. It produces a 1 at its output only if all inputs are 1.
- C. It produces a 0 at its output if some but not all inputs are 1.
- D. It produces a 0 at its output only if all inputs are 1.

E7A08. What logical operation does an OR gate perform?
- A. It produces a 1 at its output if any input is 1.
- B. It produces a 0 at its output if all inputs are 1.
- C. It produces a 0 at its output if some but not all inputs are 1.
- D. It produces a 1 at its output if all inputs are 0.

E7A09. What logical operation is performed by a two-input exclusive NOR gate?
- A. It produces a 0 at its output only if all inputs are 0.
- B. It produces a 1 at its output only if all inputs are 1.
- C. It produces a 0 at its output if one and only one of its inputs is 1.
- D. It produces a 1 at its output if one and only one input is 1.

E7A10. What is a truth table?
- A. A list of inputs and corresponding outputs for an op-amp.
- B. A list of inputs and corresponding outputs for a digital device.
- C. A diagram showing logic states when the digital gate output is true.
- D. A table of logic symbols that indicate the logic states of an op-amp.

E7A11. What does “positive logic” mean in reference to logic devices?
- A. The logic devices have high noise immunity.
- B. High voltage represents a 1, low voltage a 0.
- C. The logic circuit is in the “true” condition.
- D. 1s and 0s are defined as different positive voltage levels.

Answer Key

E7A01: C	E7A02: A	E7A03: B	E7A04: A
E7A05: D	E7A06: A	E7A07: D	E7A08: A
E7A09: C	E7A10: B	E7A11: B	